

GAN-Loc: Empowering Indoor Localization for Unknown Areas via Generative Fingerprint Map

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Abstract—As indoor localization becomes a necessity to provide intelligent location-based services for in-building users, fingerprint-based positioning has been widely adopted in numerous Wi-Fi-equipped devices. However, its reliance on extensive offline site surveys limits its application in unexplored environments without prior fingerprint sampling. To address the challenge, we propose *GAN-Loc*, a novel generative fingerprint-based framework. We extract the underlying correlation between location and radio signal features, empowering indoor localization in un-/under-explored areas, including unknown data points, newly deployed APs, or unexplored sites. It involves: 1) decomposing into a signal feature map for each AP perspective; 2) learning with a set of points and their associated signal strength data; 3) generating and integrating synthetic radio fingerprints; and 4) employing them into some existing localization algorithms. We evaluated *GAN-Loc* with extensive real-world RSSI measurements in seven different indoor places. *GAN-Loc* achieves the localization accuracy of up to 1.21m, 1.29m, and 1.28m for Wi-Fi, Zigbee, and BLE, respectively, compared to the accuracy of 1.47m, 1.58m, and 1.22m using an ideal ground-truth map, which is unachievable without site survey in unknown sites.

I. INTRODUCTION

With the rapid advancements in technology and innovation, indoor user positioning has become one of the core features for business marketing, smart home/industry automation, facility security, life-safety operations, and assistive technology. Recent artificial intelligence (AI)-based technology brings out innovative applications, leading to *indoor intelligence*. To accelerate its development, wireless fingerprints offered by nearby devices encapsulate underlying temporal and spatial features for users' positioning or pose estimations [1], [2]. This wireless fingerprint-based localization has been extensively researched to achieve high-fidelity indoor positioning.

However, in indoor environments, where obstructions such as walls and furnitures are common, it is essential to use wireless radios that do not rely on Line-of-Sight (LoS). Although some Global Navigation Satellite System (GNSS) or Radio Frequency (RF)-based localization techniques, e.g.,

mmWave, can provide accurate estimations, they require additional equipment and LoS conditions [3]. In contrast, wireless radios such as Wi-Fi, BLE, and ZigBee are highly applicable especially in indoor environments due to their broad deployments and non-LoS usability. Indoor localization with these wireless radios can easily utilize unique radio signal features captured by a device carried by a user to create a characteristic fingerprint associated with any LoS or non-LoS specific locations. Here, a radio map of signal strengths is constructed for each one-on-one reference point indoors during an *offline site survey*. When a user is to be positioned, the closest pre-constructed fingerprint is identified based on real-time radio signal measurements at execution time. However, the scalability of fingerprint-based localization is limited since it cannot be directly applied to unexplored points or sites.

To mitigate the burden on offline site surveys, some research expands upon radio map based on generative learning. These approaches conduct data augmentation by generating a synthetic radio fingerprint map using Variational Autoencoders (VAE) [4], Conditional Generative Adversarial Network with Auxiliary Classifier (AC-GAN) [5], or a combination of deep learning models of CNN, LSTM, and CGAN [1]. Although the resolution problem can be alleviated via data augmentation, the inherent site-dependent fingerprint mapping still remains to be an unresolved problem, especially for unexplored areas.

Here, we raise some fundamental questions – *is it possible to decompose the radio signal traces collected at a certain place into some generic key features around each Access Point (AP)? If so, is it possible to re-compose them for localizing unexplored points or sites, with a different AP distribution or even at a totally new site?* Recent advances in deep learning suggest desirable guidance for extracting essential features from collected traces. In particular, Generative Adversarial Network (GAN), which is one of the commonly used generative models, can offer powerful assistance in efficiently extracting the features and approximating an unknown indoor infrastructure by generating a synthetic radio map distribution.

In this paper, we revisit the indoor localization problem by discovering a new perspective. The premise is that the underlying patterns of radio signal propagation within indoor spaces fall into a range of parameters or hidden metric spaces

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that can be learned. They can be applied to previously unseen areas such that a site survey can be skipped, enabling localization without the need for time-consuming and labor-intensive site survey anymore. Instead of presenting a new localization algorithm, we aim to advance fingerprint-based localization and positioning by generating a synthetic site-independent radio map for *under-explored* or *un-explored* areas.

We introduce *GAN-Loc*, a generative model-inspired radio map inference framework for unsupervised indoor localization. *GAN-Loc* employs conditional GANs [5], [6] to generate higher-resolution radio maps for areas with existing coverage or entirely new radio maps for indoor rooms or areas that have not been profiled for localization. We use coarse-grained Received Signal Strengths Indicator (RSSI) rather than fine-grained Channel State Information (CSI) since RSSI is more commonly available from low-end to high-end wireless devices, e.g., Wi-Fi, Bluetooth, and ZigBee. *GAN-Loc* takes the relative coordinates from APs as input along with their associated RSSI values. *GAN-Loc* processes a series of decomposition, learning, and generation to create an authentic-like radio map and then performs localization on top of the generated radio map. In particular, *GAN-Loc* engages two GAN variants of CGAN [6] and AC-GAN [5] to extract common and general features from the association between radio signal features and location coordinates from APs.

This work is the first to empower fingerprint-based indoor localization for unexplored areas based on generative models. The main contributions can be summarized as follows:

- (i) We explore a novel generative model-inspired radio map inference framework, *GAN-Loc*, for unsupervised indoor localization, significantly reducing the reliance on site surveys.
- (ii) We offer an orthogonal approach to integrate synthetic radio map generation with the existing localization algorithms. *GAN-Loc* facilitates or empowers them for inferring the position of users in any un-trained or under-trained areas.
- (iii) We conduct extensive experiments based on real-world RSSI traces in seven in-building places with various wireless radios. We demonstrated that *GAN-Loc* in an unknown area provides localization accuracy of 1.21m, 1.29m, and 1.28m compared to the ground-truth maps with the accuracy of 1.47m, 1.58m, and 1.22m using Wi-Fi, Zigbee, and BLE, respectively. We also reduced the distance error by 34.2% at the same place by augmenting unrevealed fingerprints.

II. RELATED WORK

With the growth of wireless devices, many researchers have investigated fingerprinting localization using the signal strength of RSSI [2], [7]. *RADAR* [7]—the pioneering work for fingerprint-based localization—stores the location-tagged signal fingerprints in the offline phase. Then, it estimates the user location by finding some best-matching fingerprints with online measurements based on Nearest Neighbor (NN) algorithms in run-time. However, the site survey during the offline phase has proved that more efforts on data collection to populate the fingerprints in the radio map enable the

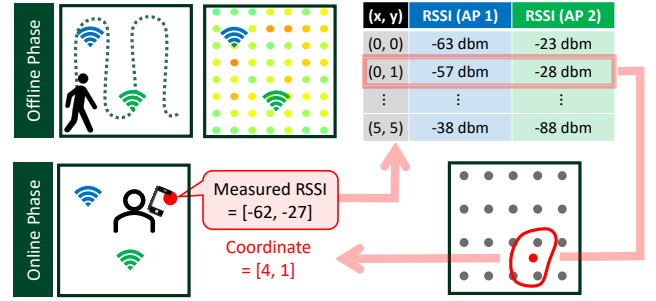


Fig. 1. Radio fingerprint-based indoor localization

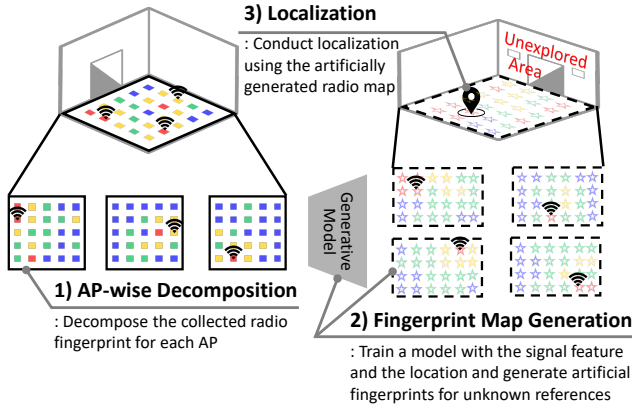
more precise localization [8]. To tackle the labor- and time-intensiveness of offline phase, data collection using robots is proposed to reduce the labor force [9]. The limitation of robots is that only free space is reachable, and it is hard to measure the accurate location of the robot. Moreover, as a critical issue, most of the works on indoor localization evaluate some particular points only with 0.07 to 1.79 per 1 m^2 .

As machine learning-driven techniques achieve high performance in regression, deep learning-driven localization [10] has been recently investigated by predicting the radio features rather than manual collection. More closely related to our work, some studies exploit GAN [1], [11] or autoencoder [12] to assist site survey, thereby improving the localization accuracy. In particular, *Selective GAN* [11] adopts Vanilla GAN to generate unlabeled fingerprints. However, since the existing approaches imitate the unlabeled RSSI vectors, they cannot specify where to generate and continuously produce fingerprints at a certain area, which involves more samples in training stage. This imbalance problem through uncontrolled fingerprint generation incurs a biased localization towards the fingerprint-crowded points. Moreover, they focus only on data augmentation at the same site. However, with new additions or removals of APs, these already-constructed maps cannot directly be applied even to the same areas. Moreover, it becomes more infeasible when applied into new areas.

III. SYSTEM MODEL

Typically, a fingerprint-based localization is conducted with two phases: offline and online phases [7], as illustrated in Fig. 1. In the offline phase, the initial setup is launched to construct a fingerprint map—a set of the signal vectors for each site—by collecting signal values at each reference point. After the site survey, the online phase is followed to estimate the user position by comparing the real-time measurement with the fingerprint radio map and performing localization for users. Thus, the fingerprint-based localization highly depends on the radio map construction with a sufficient spatial resolution [8].

We take an approach to infer a realistic radio map based on RSSI measurements over a region of interest (RoI). While variations in signal propagation due to factors such as building structure, AP configurations, and user behavior can affect RSSI, we strategically focus on capturing high-level propagation characteristics, serving as a general model. We take a rather daring approach of fingerprint map generation for unknown sites, which has been a type of otherwise impossible

Fig. 2. Overview of *GAN-Loc* with fingerprint map inference

localization. We then leverage the constructed fingerprint map to estimate the position for the measured values during the test phase. It is designed to be orthogonal to any fingerprint-based localization methods across machine learning approaches, probabilistic methodologies (e.g., Bayesian), or averaging algorithms (e.g., NN, kNN, and the weighted kNN).

We present *GAN-Loc*, a simple yet efficient radio map inference framework without requiring a prior radio fingerprint collection for the online test phase. Our goal is to achieve indoor localization over the *under-explored* area with insufficient fingerprint samples or totally *un-explored* areas. Given the pre-collected radio maps from a certain area, we perform a series of procedures of 1) decomposition; 2) learning and generation; and 3) localization to predict the location of the users in unknown environments as in Fig. 2.

We address the problem of fingerprint-based indoor localization for *under-* or *un-explored* indoor areas as follows (Fig. 3):

- (i) *Unexplored* data points in the same site: augmenting the uncollected points to make a map with a higher resolution
- (ii) *Unexplored* APs in the same site: inferring signal values under a newly deployed or migrated AP layout
- (iii) *Unexplored* sites: fully generating a whole synthetic fingerprint map for new places

IV. GENERATIVE FINGERPRINT MAP

We design a GAN architecture to extract the generic key features of radio signal propagation from each AP by training the distribution between the user location and the radio fingerprint at the location. We then apply the extracted feature model to predict the radio signal values at the unknown points. By empowering with GAN, we generate a denser radio map or a complete artificial radio map unless otherwise impossible for localization. Based on radio map inference, we are capable of imitating realistic fingerprints for target points or augmenting the partially acquired fingerprints by integrating a set of synthesized signal strength values from multiple APs.

In particular, as unbalanced fingerprints can overestimate on dense areas [11], generative models with location-related conditions by specifying a target area are necessary to generate a robust unbiased map. To generate authentic-like data under a specific condition indicating the location, we can employ

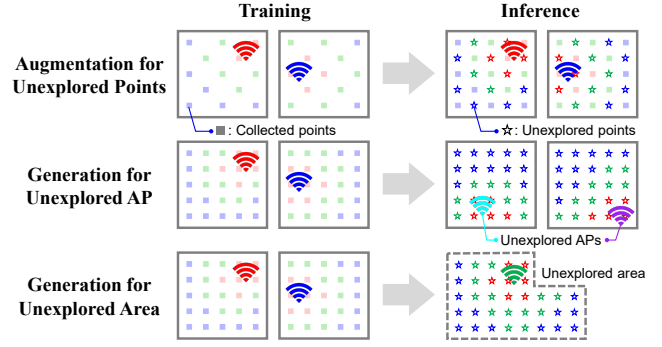


Fig. 3. Three main use cases of unexplored indoor areas

any variants of conditional GANs. Different from the vanilla GAN, where the possible data is indiscriminately produced, the conditional GAN aims to limit the data considering given conditions. In order for a user to be positioned in any place, fingerprints need to be dense enough to cover the area of interest. As the location-related information is fed as a condition, we obtain a radio fingerprint map estimate for the target area.

We embed two conditional GANs: 1) CGAN [6] for unexplored data points and unexplored APs in same sites: these typical CGANs generate samples closely resembling trained data, which might be effective for augmenting and mimicking fingerprint patterns within known places; and 2) AC-GAN [5] for unexplored areas: in contrast to traditional CGANs, we further employ AC-GAN that incorporates class predictions, preventing the model from being trapped beyond the seen data and thus enhancing its adaptability to unexplored sites.

A. AP-wise Map Decomposition

A fingerprint of each reference point is denoted with a vector of radio signal strengths as follows:

$$X(i) = \{s_1, s_2, \dots, s_{N_{ap}}\} \text{ for } i \in \{1, 2, \dots, N_{data}\} \quad (1)$$

where $X(i)$ denotes a fingerprint of the i -th point, s_j is the signal strength from AP j , N_{ap} is the number of available APs, and m is the number of data points. To extract the common attributes of radio from distinct APs, we convert each single radio map of multiple APs into multiple maps as follows:

$$X_j(i) = s_j \text{ for } j \in \{1, 2, \dots, N_{ap}\} \quad (2)$$

The number of points increases with a factor of N_{ap} , and the amount of data in each entry is reduced by a factor of N_{ap} .

We regard a single signal value at a reference point as a single data. It has widely been observed that the distance between the target and the transmitter AP is an important factor, as disclosed in propagation models such as the log-distance path loss [13]. Beyond the distance measure, we also use the positioning angle from the AP to distinguish the unique point for the distinct reference point. In our model, the fingerprint for the i -th point from AP j is defined as follows:

$$X_j(d_i, a_i) = s_j \text{ for } j \in \{1, 2, \dots, N_{ap}\} \quad (3)$$

where d_i and a_i are the distance and the angle from AP j , respectively. These measures have different ranges, and thus they are normalized from 0 to 1 and fed into the condition.

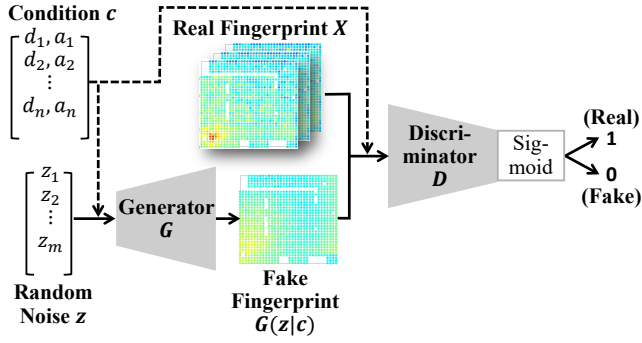


Fig. 4. Radio fingerprint generation based on a CGAN with the condition of relative distance (d_i) and angle coordinates (a_i) from a distinct AP

B. Fingerprint Map Generation

We train a conditional GAN with RSSI data and the reference point at which they are measured as the condition as in Fig. 4. Based on the trained model, we generate authentic-like signal strength values for unknown points. For the same sites, we can achieve more accurate localization by augmenting the fingerprint. In the case of new sites, we generate a completely new radio map over the whole area. Through the artificial radio map inference without prior site survey for acquiring sufficient fingerprint samples, *GAN-Loc* is used as a core primitive to be extended and combined with localization.

Learning process. A conditional GAN consists of two hostile networks, generator G and discriminator D . Generator G learns the distribution of real fingerprints x_{real} and fabricates fingerprint $G(z|c)$ from the random noise samples z under condition c . We train signal strength data given the condition set of the relative distance d and the angle a from an AP to a reference point. As the distribution of fake data $x_{fake} = G(z|c)$ becomes similar to the real distribution, the generator succeeds in deluding the discriminator, which means $D(G(z|c))$ gets close to 1. On the other hand, discriminator D distinguishes fake one $G(z|c)$ from real one ($x|c$). The goal of the discriminator is to estimate the fake fingerprint $D(G(z|c))$ closer to 0 and the real fingerprint $D(x|c)$ closer to 1. Therefore, the objective function of the conditional GAN is defined based on a min-max game as follows:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}(x_{real})} [\log D(x|c)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z|c)))] \quad (4)$$

where the distribution of the real and the fake fingerprints are $p_{data}(x_{real})$ and $z \sim p_z(z)$, respectively. We alternate the training process of the discriminator and the generator until both networks become converged.

When generator G is trapped in an overfitted *mode* that succeeded in deceiving discriminator D , it repetitively outputs monotonous samples. To prevent this *mode collapse* problem, we also exploit AC-GAN [5], which trains not only the sample loss, but also the class loss. *GAN-Loc* with AC-GAN has two classifiers after the discriminator: 1) sample classifier to discriminate the real data as 1 and the fake data as 0 as in CGAN; and 2) additional class classifier to predict the conditions of relative distance and angle. The discriminator

aims to minimize both sample loss and class loss, whereas the generator minimizes class loss, but maximizes the sample loss. As AC-GAN also trains fingerprints to be predicted as the corresponding conditions, it has advantages in escaping from the monotonous mode and in generating diverse outputs. It means that we can benefit from AC-GAN for expanding to new places, while the vanilla CGAN can tightly be fitted into augmenting on a target area or generating a similar target area.

Generation process. After saturation, we feed random noise samples under the condition of a location of interest into the trained generator and obtain an imitation. We repeatedly run the generation process over every data point on a target fingerprint map to construct. Even with the same noise inputs, we can generate a different signal strength for a distinct position due to the different representations of the condition.

The generated map for each AP is merged into a map of multiple signal strength vectors. It can be utilized as the map of multiple APs. For data augmentation, we fully use the real data and fill the unknown points with the generated data. With a new AP distribution or at an unexplored area, we generate a whole set of fingerprints over the area, which has not been revealed. If only some of the AP distributions are varied, we generate only the partial distribution for newly deployed APs.

C. Localization Coupled with Inferred Map

In the online phase for a different test environment, we conduct localization for given new radio measurements and the generated fingerprint map. Given an inferred map, we first measure the signal strengths instantly at a certain place where the user is located to find out the original position. Then, we estimate the probable location of a user by searching for the best matching fingerprints from the generated map, together with one of any fingerprinting-based localization algorithms.

V. EXPERIMENTS

We validated our approach over seven real-world places from (i) our own Wi-Fi testbeds and (ii) public datasets [14] with various wireless radios of Wi-Fi, BLE, and Zigbee.

(i) Our own dataset. In our own testbeds as shown in Fig. 5, we collected real-world RSSI data on the floor in four in-building places over two different buildings with differently obstructed structures: 1) **Hall**: Open space of hall, using 1 to 9 APs over the RoI of $8.1 \times 8.7m^2$ with 831 data points; 2) **Lounge**: Less-obstructed space of lounge with the open wall, using 1 to 12 APs over the RoI of $6.6 \times 9.9m^2$ with 764 data points; 3) **Ofc-Obs**: Obstructed space of six offices with doors opened, using 2 to 16 APs over the RoI of $9.9 \times 9.9m^2$ with 1067 data points; and 4) **Ofc-HighObs**: Highly-obstructed space of six conference rooms with half doors closed, using 1 to 10 APs over the RoI of $6.3 \times 10.8m^2$ with 756 data points.

We used 26 different LG G Pad 3 10.1 (LG-X760) devices along with our custom Android application to collect RSSI values. We measured RSSI values 30 times consecutively with every 3 seconds over the RoI with an interval of 0.3m using *WifiManager* in Android Studio. For the APs, EFM-Networks ipTIME N104Q-i with the single antenna of 5 dBi using IEEE 802.11n of 150 Mbps over 2.4 GHz band is deployed. As we

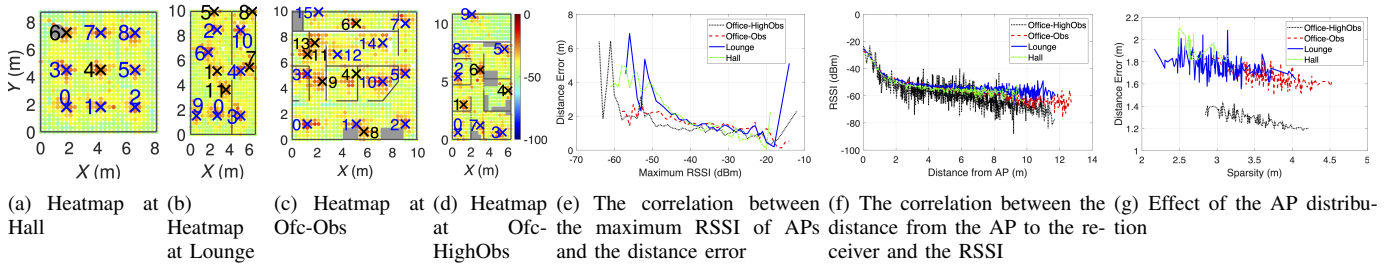


Fig. 5. The heatmaps of maximum RSSI among detectable APs (where APs are marked with 'X', and the 'X' with blue color represents the mainly used AP layout with the best accuracy) and the observations of radio features

used a low-powered transmitter only with a single antenna, we mainly used one AP per unit area of $10 m^2$. This ranged from 1 to 16 APs per place, equivalent to 0.41 to 5.14 APs per standard private office size of $28 m^2$, compared to reported densities of 0.19-0.34 APs [15] and 0.60 APs [16] per $28 m^2$.

(ii) Public datasets. We used RSSI values across Wi-Fi, Zigbee, and BLE collected from three places [14]: 1) **Public-LowObs**: Meeting room using 3 APs over the RoI of $6.0 \times 5.5 m^2$ with 59 data points; 2) **Public-MediumObs**: Computer lab using 3 APs over the RoI of $10.8 \times 7.3 m^2$ with 56 data points; and 3) **Public-HighObs**: Meeting room with high noise using 3 APs over the RoI of $5.8 \times 5.3 m^2$ with 33 data points.

Model configurations. We design GANs with one hidden layer of 128 neurons using ReLU activation function. The Sigmoid activation function is used for the output layer, and the Adam optimizer is adopted with a learning rate of 0.001. As the input of the generator network, a random noise of 100 uniformly distributed variables over $U[-1, 1]$ is sampled.

Measures. We quantified localization performance using two metrics: 1) accuracy in meters, which is calculated to be the averaged distance error between the ground-truth position and its predicted position; and 2) precision in percentage within a distance error range of $1.5 m$ to be the portion of test points as correct. We showed the performance of the best AP distribution, which has the highest accuracy in ground-truth data. As the RSSI varies over time, we report the averaged performance of five different RSSI measurements for reliable verification for testing. The RSSI values out of range from -100 dbm to 0 dbm are regarded as abnormal measurements and not used for both training and testing.

Localization algorithms. We verified our scheme using three localization matching algorithms: 1) kNN; 2) wkNN; and 3) DNN. Since *GAN-Loc* can be used as a basis of the underlying fingerprint map, other fingerprint-based localization algorithms can easily be coupled. We mainly evaluated using wkNN, which showed the best performance among them based on real ground-truth data. Regarding the parameter k in kNN and wkNN, the five percent of total data is used for k . For DNN, the raw radio signal strength vector is used as the input data, while the coordinates of (x, y) is used as the output data.

Baselines. We conducted a comparative analysis with wkNN against some baseline models for signal value prediction, including ML-based or non-ML-based models: 1) *Upper*: the ideal baseline where all of the ground-truths for reference points are used with the highest spatial resolution with a grid

size of $0.3 \times 0.3 m^2$; 2) *Unaug.*: an unaugmented real map with the spatial resolution with a grid size of $1.5 \times 1.8 m^2$; 3) *AVG* [17]: to estimate the RSSI of unknown points by kNN averaging; 4) *Selective GAN* [11]: to augment the signal strength vectors by GAN and then label the generated vectors with the location using DNN; 5) *Log-Dist Propagation* [13]: log-distance path-loss model without a random noise; 6) *ITU Propagation* [18]: propagation model for indoor attenuation; and 7) *ROCRSSI* [19]: to estimate the location by comparing its signal strength among some selected anchor nodes.

A. Observations in Real-World Signal Data

Based on the insights from Sec. V-A, we validated *GAN-Loc* in three different settings: 1) data augmentation of unknown points in the same site (Sec. V-B); 2) map generation for unknown APs in the same site (Sec. V-C); and 3) map generation for a totally new site (Sec. V-D). As we go from setting 1) to 3), localization becomes harder due to the increased uncertainty from the lack of environmental information.

The significance of AP-perspective distance. Since the noise or the distortion of radio signals differs across APs, capturing propagation behaviors from an AP's perspective yields better radio feature extraction. As a closer distance incurs a higher signal strength and gets less-affected by the signal strength, it results in better localization (Figs. 5(a)-5(d) and Fig. 5(e)). In Fig. 5(f), as a reference point is at a region not far from AP, e.g., within $2 m$, the RSSI value keeps decreasing with nearly 20 dbm. Conversely, the points located farther from APs reach almost the same signal strength drop spanning a $10 m$ distance. Remarkably, the number of strongly connected transmitters may be a more important factor than the total number of detectable APs for better localization. This insight is also shown in Fig. 5(g), where sparser AP distribution contributes to improved localization.

Effect the obstruction level. We can achieve better map generation in less obstructed places, whereas better localization in more obstructed places. As in Fig. 5(f), it is more difficult to predict the location due to the noise deviations in highly obstructed rooms, leading to larger gaps when compared to the ideal ground-truth map. However, as in Figs. 5(e) and 5(g), a highly obstructed place outperforms other places in terms of absolute distance error. Although the result may sound counter-intuitive at first, the obstacles or construction materials drastically degrade the signal, and thus a distinctive

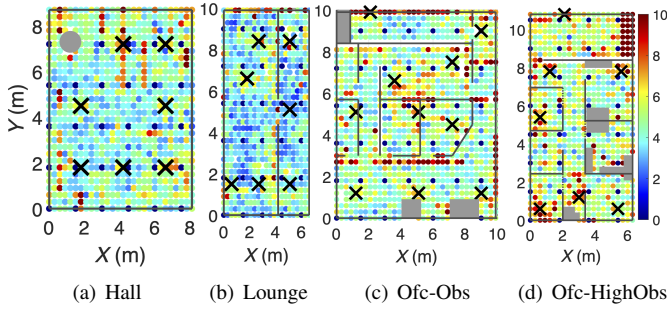


Fig. 6. RSSI error heatmaps in data augmentation

TABLE I
LOCALIZATION PERFORMANCE WITH DATA AUGMENTATION

		Accuracy (m) / Precision (%)	
Train & Test	Unaug.	GAN-Loc	
		CGAN	AC-GAN
Hall	2.82 / 32.3	2.05 / 49.2	2.05 / 49.2
Lounge	2.36 / 40.3	1.69 / 53.8	1.72 / 52.5
Ofc-Obs	2.40 / 37.2	1.65 / 53.9	1.86 / 48.5
Ofc-HighObs	2.02 / 44.1	1.33 / 66.3	1.36 / 65.3

signal environment characterized by specific obstructions can be a rather helpful asset to distinguish discrete spaces.

B. Augmentation for Unexplored Points in the Same Site

First, as the more number of fingerprints achieves better localization [8], we verified *GAN-Loc* under the same sites with data augmentation. Although the model may learn and overfit the data, the points to be predicted are closely linked to the distribution of the training data in the same sites, indicating that sufficient training rather assures successful localization.

GAN-Loc for data augmentation. We first present the RSSI error compared against real signal strengths in Fig. 6. Given the size of $1.5 \times 1.8m^2$ grid, we discover most of the hotspots with a shade difference. If we only consider the distance from the APs similar to propagation models, we could not learn the distinct in-building structure, all points in the same distance inevitably have the same signal strength. It is necessary to consider the direction of the signal to represent a distinct point. If we more closely dive into the fake fingerprints in Table I, we show a great improvement up to $0.77m$ by filling out unknown points, enhancing the spatial resolution for finer localization. Significantly, *GAN-Loc* consistently demonstrates superior performance when employing CGAN than AC-GAN. This is due to the objectives, where AC-GAN focuses on discovering diverse new fingerprints, while CGAN tends to fit into the fingerprints used in training. CGAN thus proves more effective at generating enhanced fingerprints for data augmentation within the same training and testing environments.

When we compare the distance error using the real map and the augmented map in Fig. 7, a real map (Fig. 7(a)) lacks data, resulting in the weak points extensively. However, if we augment data using *GAN-Loc*, we can fill up the unknown points by fabricating the real data and succeed in achieving high and broad localization. In particular, as we pointed out in Sec. V-A, the precise localization is more likely to be

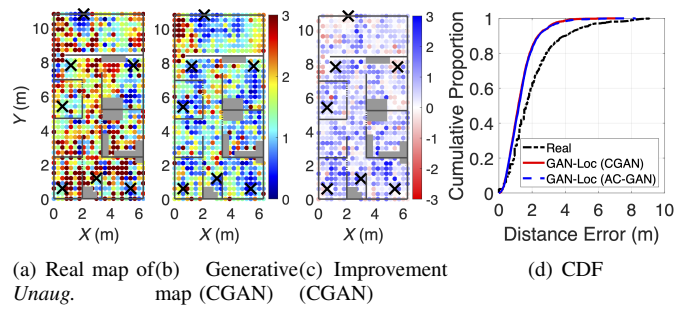


Fig. 7. Localization heatmaps and cumulative distribution of given real map and augmented map of Ofc-HighObs for unexplored data points

near the APs. Meanwhile, there are also some weak points in Fig. 7(b). The points around (0,10) estimate the poor localization because there is only one meaningful AP due to the blockage of the wall at $Y = 8$ from other APs. We analyze localization improvement (Fig. 7(c)) and a cumulative distribution (Fig. 7(d)) of the real map and the augmented map. Localization without augmentation has a more serious long tail problem, where 18.8% of the data points show a distance error larger than $3m$, compared to only 5.0% in our augmented map.

Different resolution of training data. We investigated the effect of data augmentation with regard to the resolution of train map from $0.6 \times 0.6m^2$ to $2.1 \times 2.1m^2$ in Fig. 8. In the case of *Selective GAN*, it only fabricates the signal strength vectors and thus cannot generate the fingerprints for a specified area. Since the vanilla GAN without any constraints generates towards the majority groups, it may only produce somewhat biased and similar fingerprints. Although *Selective GAN* additionally narrows down the number of fingerprints proportional to the unit area to prevent excessive data cohesion, the fingerprint imbalance in the data-less area does not have a meaningful effect on localization. When the training data is densely distributed, AVG is sufficient to estimate the unknown points. However, when the train data becomes sparse, localization only using the limited number of training data incurs high localization failure. Consequently, the kNN averaging severely degrades due to the lack of samples, whereas our GAN-based inference keeps relatively stable distance accuracy. Based on this result, we mainly use the granularity of $1.5 \times 1.8m^2$ to evaluate the data augmentation, where our prediction model turns out to be effective.

Comparison. Compared to the existing data augmentation approaches in Fig. 9, all methods also show some improvement compared to *Unaug.*, which does not apply any augmentation. We applied the estimated parameters for propagation models using training data, and they seem more powerful in free spaces. However, since propagation models are simple empirical mathematical formulations, they highly collapse in the obstructed places. On the other hand, AVG and *Selective GAN* give a better improvement in obstructed places. However, AVG based on kNN highly depends on reasonable neighbor data points. *Selective GAN* is limited to specify into a target area, and thus it shows a higher distance error with a factor of 1.20, 1.18, 1.29, and 1.38, compared to *GAN-Loc* (CGAN).

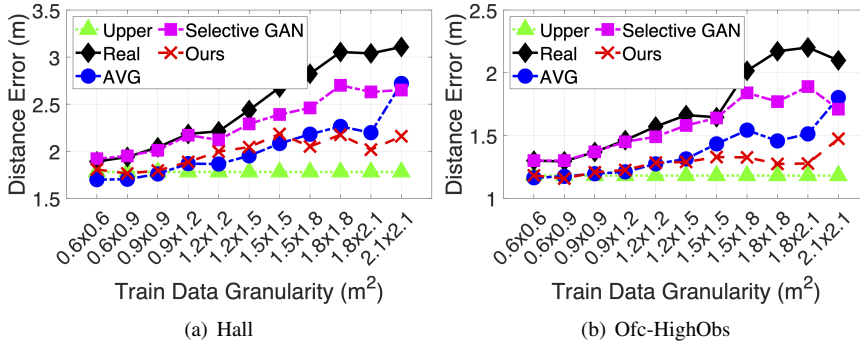


Fig. 8. Localization accuracy with regard to the resolution of train data for unexplored points (CGAN)

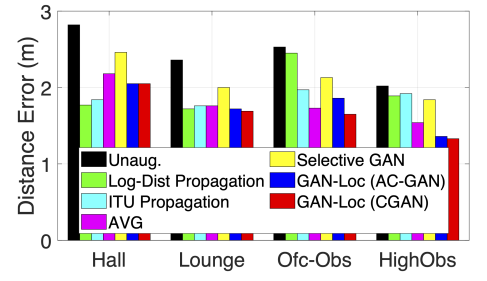


Fig. 9. Comparison to baselines with unexplored points augmentation

TABLE II
LOCALIZATION PERFORMANCE IN DIFFERENT AP DISTRIBUTIONS

Train & Test	Upper	GAN-Loc	
		CGAN	AC-GAN
Hall	1.78 / 51.9	1.81 / 57.0	1.92 / 52.2
Lounge	1.56 / 55.3	1.76 / 56.1	1.78 / 50.5
Ofc-Obs	1.48 / 58.0	1.56 / 55.3	1.80 / 44.4
Ofc-HighObs	1.18 / 73.9	1.66 / 55.4	1.91 / 49.7

C. Unexplored APs in the Same Site

We investigate the radio map in the same place but with differently deployed APs. Although APs are in the same place, the map for new or moved APs has to be fully generated.

GAN-Loc for unexplored APs. We evaluated our model in a highly-obstructed place compared to the ideal radio map, which consists of unknown real fingerprints. As shown in Table II, we achieved almost similar performance compared to the ground-truth RSSI fingerprints except for the boundary area. The reason is that the train and the test environments are identical, and consequently the propagation of each AP tends to be similar. The precision less than or equal to $1.5m$ takes charge of 55.4% over a highly-obstructed place, as denoted in Table II. Regardless of structures, both accuracy and precision keep the trivial gap with the upper-bound, although we do not have any information about the fingerprints of newly deployed APs. Similar to the results of data augmentation from Table I, since CGAN takes some benefits from memorization effect, CGAN enables better localization in the same place.

Remark 1 (GAN-Loc in under-explored areas): If some pre-collected data are given, it is better to augment unknown points. GAN-Loc enables reasonable data augmentation, whereas data augmentation using basic averaging or existing localization is not feasible under the lack of data.

D. Unexplored Sites

Different from previous experiments, in unexplored areas, we have to generate a completely new radio map without location-related information except for the AP locations.

GAN-Loc for unexplored sites. We summarized the localization accuracy and precision by varying the training and testing places in Table III. It should be noted that *Upper* is the ideal performance with the ground-truth fingerprints, which is not reachable in real-world tests. In obstructed places, it would

be better to train the area with obstacles such as Ofc-HighObs to Ofc-Obs or vice versa. In most cases, AC-GAN shows better localization when compared to CGAN. It is because AC-GAN has a strength for producing a more diverse range of outputs. While this suggests that GAN-Loc with AC-GAN is generally more compatible with new places, the difference in performance between CGAN and AC-GAN is relatively minor.

Different number of APs. We varied the number of deployed APs in Fig. 10. With more APs, the amount of data in each fingerprint certainly increases, and thus we can achieve better localization. However, once a certain number of APs are reached, any additional increase in performance is negligible.

Comparison. We compared our GAN-based map generation to non-ML classic approaches such as a log-distance path-loss propagation model [20], ITU propagation model [18], and *ROCRSSI* [19] in Fig. 11(a). The distance error highly varies with respect to the path-loss exponent (PLE). We showed the results using the estimated PLE of 1.6, 2.0, 3.0, and 4.0 [21] for the log-distance propagation and using the pair of loss coefficient and penetration loss factor of (20, 4), (22, 6), (28, 4), and (30, 5) for the ITU model, in Hall, Lounge, Ofc-Obs, and Ofc-HighObs, respectively. It should be noted that all factors need to be estimated via the underlying network measurements, which is almost impossible to find suitable values without a site survey, while we do not need the site survey in new environments. *ROCRSSI* estimates the location by comparing its signal strength among some selected anchor nodes, whereas GAN-Loc does not use any signal information from the unexplored areas. Without any prior data collection for the new sites, GAN-Loc can reduce the localization accuracy more than *ROCRSSI* with a factor of 0.64, 0.56, 0.59, and 0.57 in each place. Compared to the *Upper*, where the whole ground-truth fingerprint map is known, although we do not outperform the ideal cases, we keep a fairly competitive performance without any pre-collected fingerprints in tests.

Orthogonality to localization algorithms. We investigated the distance accuracy using various localization algorithms such as DNN, kNN, and wkNN in Fig. 11(b). As GAN-Loc solves the problem of the offline phase that generates the radio map, it is orthogonal to the online phase algorithms that estimate the location from the radio map. Our generated map using GAN-Loc provides comparable localization error regardless of indoor localization algorithms.

TABLE III
LOCALIZATION ACCURACY AND PRECISION IN NEW ENVIRONMENTS

Train \ Test	Hall		Lounge		Ofc-Obs		Ofc-HighObs	
	CGAN	AC-GAN	CGAN	AC-GAN	CGAN	AC-GAN	CGAN	AC-GAN
Hall	—	—	1.79 / 51.5	1.74 / 53.8	1.76 / 52.9	1.65 / 52.6	1.58 / 60.5	1.44 / 62.0
Lounge	1.97 / 54.2	1.87 / 55.5	—	—	1.72 / 55.1	1.68 / 55.1	1.61 / 55.9	1.53 / 58.7
Ofc-Obs	1.83 / 57.1	1.88 / 50.2	1.61 / 55.9	1.89 / 42.9	—	—	1.37 / 63.4	1.37 / 66.3
Ofc-HighObs	2.03 / 47.2	1.94 / 51.0	2.04 / 39.8	1.97 / 43.4	2.04 / 35.2	1.62 / 53.3	—	—
All Other Places	1.86 / 50.0	2.01 / 50.2	1.79 / 48.6	1.74 / 53.5	1.61 / 52.9	1.63 / 52.7	1.51 / 58.4	1.36 / 65.6
Upper	1.78 / 51.9		1.56 / 55.3		1.48 / 58.0		1.18 / 74.9	

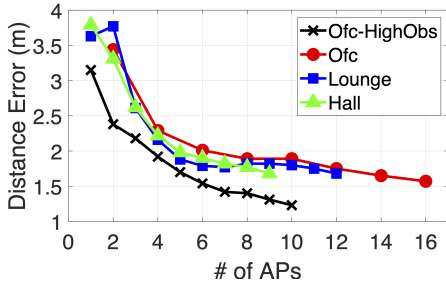
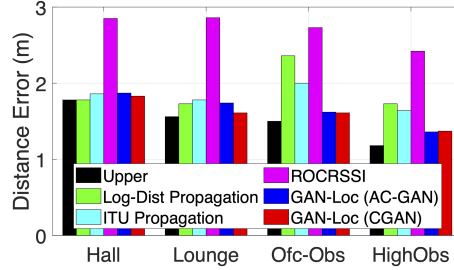
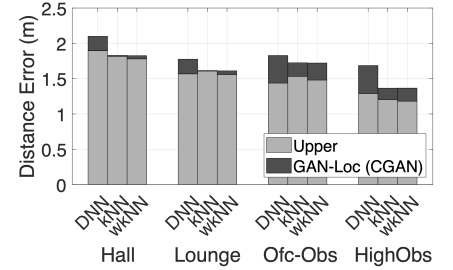


Fig. 10. Performance with respect to the number of APs for unexplored sites



(a) Comparison to baselines



(b) Localization matching algorithms

Fig. 11. Comparison to baselines and various localization algorithms and in new areas

TABLE IV
LOCALIZATION ACCURACY ACROSS VARIOUS WIRELESS RADIOS. *GAN-Loc* WAS VALIDATED ON THE GENERATED MAP WITH TWO DIFFERENT RESOLUTIONS FOR EACH TARGET PLACE BY USING ALL OTHER PLACES FOR TRAINING, WHILE *Real* WAS TESTED VIA A FIVE-FOLD CROSS-VALIDATION

	The same resolution (m) / The denser resolution to the ground-truth map (m)								
	Wi-Fi			Zigbee			BLE		
	CGAN	AC-GAN	Real	CGAN	AC-GAN	Real	CGAN	AC-GAN	Real
Public-LowObs	1.37 / 1.21	1.51 / 1.25	1.47	1.57 / 1.29	1.52 / 1.49	1.58	1.40 / 1.46	1.85 / 1.28	1.22
Public-MediumObs	1.87 / 1.87	2.53 / 1.88	<u>1.49</u>	2.45 / 1.57	2.42 / 2.47	3.41	3.01 / 1.36	2.34 / 1.77	1.80
Public-HighObs	2.00 / 1.75	2.30 / 1.72	<u>1.58</u>	2.44 / 1.83	2.62 / 1.41	3.20	1.79 / 2.22	1.93 / 1.37	1.67

Applicability into various wireless radios. We further validated *GAN-Loc* with different wireless technologies of Wi-Fi, Zigbee, and BLE using public datasets of real-world RSSI in three environments [14]. In Table IV, we validated our generated maps with two different resolutions: 1) the same resolution to the ground-truth; and 2) a denser resolution with $0.1 \times 0.1m^2$. First, the result of the map with the same resolution across various wireless radio technologies shows a similar trend in Table III, where the maps generated by *GAN-Loc* almost reach or even outperform *Real* within the less-obstructed sites (e.g., Public-LowObs in Table IV and Hall in Table III). This is due to the fact that a radio map with less noise is easier to learn and predict. As BLE, a low energy wireless radio tends to have lower RSSI signal strength and span the narrower RSSI range compared to Wi-Fi and Zigbee, *GAN-Loc* can attain a similar localization to *Real*. Interestingly, the efficacy of BLE-based localization becomes more suitable in smaller spaces due to its inherently weaker signal strength, and thus the gaps between the generated map and the *Real* map become reduced as the size of RoI decreases from Public-MediumObs, Public-LowObs, to Public-HighObs.

In the case of the map with a denser resolution, as revealed by the results of the augmentation (Sec. V-B) and the sparsity

(Fig. 5(g)), we expect to have enhanced accuracy due to the finer granularity of the map. As in Table IV, even with an equally trained model, we achieved significantly improved localization of $1.62m$ using CGAN and $1.63m$ using AC-GAN by generating a fine-grained map, compared to its original ground-truth map (*Real*) with an accuracy of $1.94m$ on average. Taking a closer look at each case, *GAN-Loc* outperforms the ground-truth map in 6 out of 9 cases, with gaps reaching up to $1.84m$. This indicates that the potential for significantly surpassing the ground-truth by offering a fine-grained map without any prior radio fingerprints for new sites.

Remark 2 (*GAN-Loc* in unexplored areas): We have verified that localization in unexplored areas is available by fabricating some radio maps across various wireless radios. *GAN-Loc* shows promising localization accuracy even at highly obstructed places, while achieving comparable localization performance to the ideal case.

VI. DISCUSSIONS AND LIMITATIONS

GAN-Loc has shown the possibility of indoor localization for unexplored areas or APs, as summarized in Table V.

Analysis on the mode collapse problem. To show the diversity of generated maps, we quantify it using the MultiScale Structural Similarity Index Measure (MS-SSIM). MS-

TABLE V
SUMMARY OF PERFORMANCE COMPARISONS IN OFC-HIGHObs

	Augment	New Site
Unaug. Upper		2.02 1.18
Log-Dist Propagation [13], [20]	1.89	1.73
ITU Propagation [18]	1.92	1.64
AVG [17]	1.54	N/A
Selective GAN [11]	1.84	N/A
ROCRSSI [19]		2.42
GAN-Loc (CGAN)	1.33	1.37
GAN-Loc (AC-GAN)	1.36	1.36

SSIM assesses image quality across multiple scales, and higher MS-SSIM values indicate a higher risk of mode collapse. The ranges of MS-SSIM values of CGAN and AC-GAN are [0.07, 0.24] and [0.03, 0.13]. While AC-GAN exhibits a relatively lower MS-SSIM value compared to CGAN, both CGAN and AC-GAN models demonstrate a small risk of mode collapse (c.f., MS-SSIM for ImageNet with AC-GAN is 0.18 [5]). This low risk contributes to stable localization performance (Table III). Notably, CGAN tends to outperform AC-GAN in data augmentation and new AP distribution cases at the same site due to a memorization effect, while AC-GAN generates more diverse samples at unexplored sites.

Failing cases. *GAN-Loc* reveals potential pitfalls of generative models and the long tail problem in localization. Failing cases occur, particularly when generating samples based on unseen conditions, such as untrained large distances. The inherent imbalance in input conditions for conditional GANs may occasionally lead to relatively lower prediction performance at the border of the RoI, as in Fig. 6(d).

Dynamic environments. As we pointed out, obstructed places are challenging to identify and represent the underlying true distribution accurately. However, even with the absence of prior building information (e.g., floors or rooms), *GAN-Loc* can produce reasonable positioning, although integrating obstacle information may further improve fingerprint quality in generative models. This empowers feasible localization without the need for extensive prior data collection.

VII. CONCLUSION

We introduced *GAN-Loc*, a generative model-inspired radio map inference framework to make fingerprint-based indoor localization feasible even in unknown areas, processing a series of decomposition, learning, and radio map generation, followed by localization on the generated map. We employed conditional GANs designed to extract common unified signal features between RSSI and position, considering the relative distance and angle from the AP. The generator produces probable fingerprints for data augmentation in known environments and constructs a whole radio map for new sites. Based on extensive experiments with real-world RSSI datasets, we validated that *GAN-Loc* not only enhances indoor localization through fingerprint augmentation but also demonstrates the inference of an unknown radio map for localization in unexplored areas. The achieved performance is comparable to ideal localization via a complete site survey.

For future work, incorporating in-building structures could enhance the accuracy by learning about surrounding obstacles. Addressing sudden network or environmental variations is crucial, as constructed radio maps may quickly become outdated, leading to poor localization. Incorporating more dynamic responses and 3D radio propagation considerations will be necessary for practical localization applications.

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